BATTERY CONTROL MODULE (BCM) SOFTWARE UPDATE

AFFECTED VEHICLES: **CERTAIN 1998-2000 MODEL YEAR RANGER ELECTRIC VEHICLES (EV) EQUIPPED WITH A LEAD ACID (PbA) TRACTION BATTERY**

OVERVIEW

THIS PROCEDURE DETAILS THE UPDATE PROCESS FOR THE BCM SOFTWARE OF THE AFFECTED EV RANGER VEHICLES ONLY. POSTAL VEHICLES DO NOT REQUIRE THIS UPDATE.

SERVICE PROCEDURE

1. NOTE: Make sure the vehicle is NOT connected to a Power Control Station (PCS) and the climate control system and all other auxiliary loads are off when performing the following procedures.

   Use the standard NGS tester cable for the following procedure. Do NOT use the EEC reprogramming cable. Using the EEC reprogramming cable will cause the update to fail.

   Insert the Field Update Card into the New Generation Star (NGS) Tester
2. Connect the NGS Tester to the data link connector (DLC) located under the steering column.
3. Wait 15 seconds.
4. Turn the ignition key to the ON (RUN) position. Do NOT go to the START position. The following screen will appear:



(PbA ONLY) (VER 3.3.1)
(BOTH) (VER 20.1)
(NIMH ONLY) (VER 7.1)

01B76A

5. NOTE: Even if the XL5F-10B687-BH version is already installed, continue with this step.

   Press TRIGGER to continue. The following screen will appear:



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6. NOTE: Do NOT turn the ignition key to the OFF or START position or remove the NGS data link cable while the BCM software is being updated. It is normal for instrument cluster indicators to be in a rest position and cluster lamps to illuminate during the update.

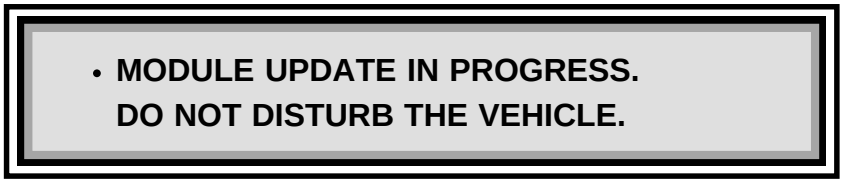
   Press TRIGGER. The following screen will appear:



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7. NOTE: **"WARNING: YOU ARE ABOUT TO OVERWRITE CURRENT SOFTWARE WITH A NEWER VERSION"** appears after pressing TRIGGER.

   Press TRIGGER. The following screen will appear:



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   Allow the update to complete (approximately 16-18 minutes). A status screen will be displayed during the update. The screen will display "UPDATE COMPLETED SUCCESSFULLY" after the update has been completed.

   If the screen displays "UPDATE FAILED" check the NGS connections and back out of the update screen using the TRIGGER button to return to the main menu. Turn the ignition key to the OFF position. Disconnect the auxiliary battery negative cable. Wait two minutes, reinstall auxiliary battery negative cable and repeat steps 1 through 7.

   If the screen still displays "UPDATE FAILED", disconnect the auxiliary battery negative cable and disconnect the IAA connector for approximately two minutes. Reinstall auxiliary battery negative cable and leave the IAA connector disconnected. Repeat steps 1 through 7.

   NOTE: It may be necessary to repeat this process several times before successfully completing the update. After the update is complete, connect the IAA connector.

8. Press TRIGGER to return to main menu.
9. With the ignition switch still in the ON position, remove the Field Update Card and insert the EV Ranger NGS Diagnostic Card.
10. Reset BCM Historic data and state of charge (SOC).

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11. Retrieve and clear, if necessary, continuous Diagnostic Trouble Codes (DTC) from the BCM, TIM and the IAA module.
12. Install a BCM update sticker as follows:
    a) Obtain and prepare the Authorized Modifications label. Write the modification completed, your P&A code, date of repair and the owner notification program number as shown. See Figure 1.
    b) Use isopropyl alcohol and a clean cloth to clean the left-hand door jam.
    c) Wait for the alcohol to dry.
    d) Peel the backing off of the BCM update sticker and install the sticker.

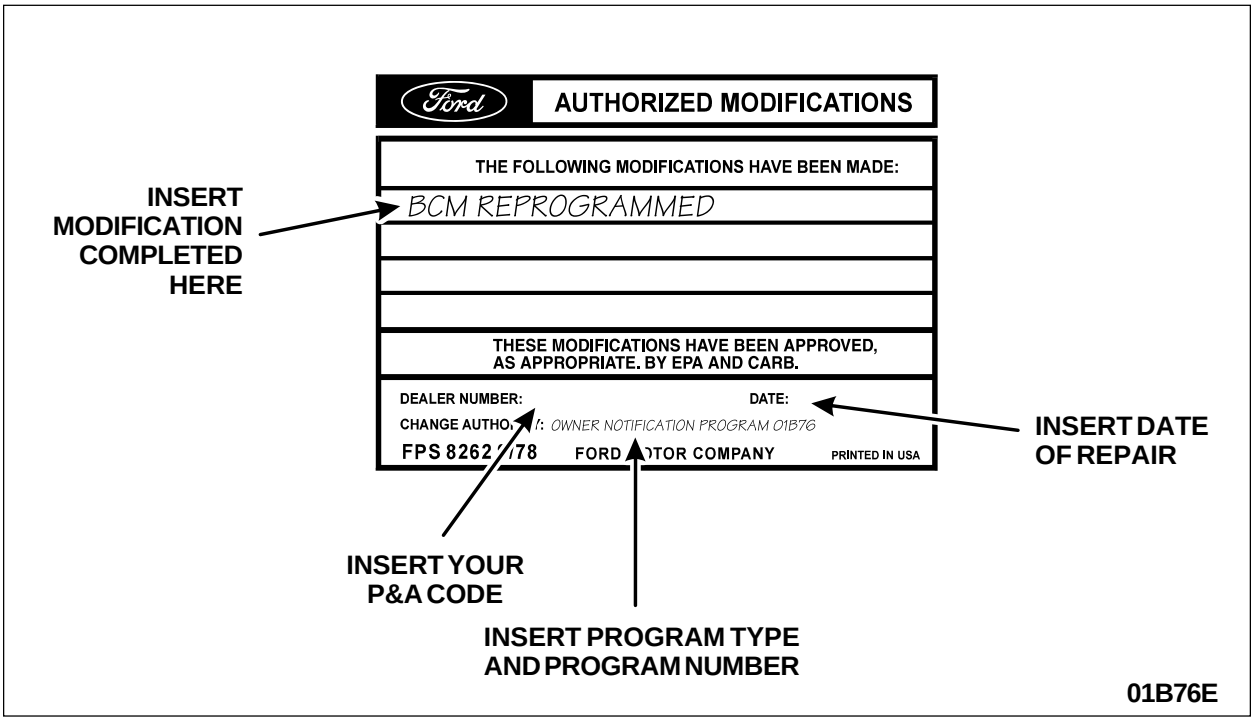


FIGURE 1

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